

Take-Home Task Breakdown

Duty Doctor Roster — Param Healthcare

1. What You're Building

A web application that **auto-generates a monthly hospital duty roster** for 6 doctors across 5 daily shifts, respecting a long list of scheduling constraints — with an editable UI backed by a real database, and manual-override support that survives re-generation.

2. Business Context

A hospital emergency department needs a doctor on duty at all times, across 5 overlapping shifts a day. This is currently a manual process (Excel / WhatsApp-style coordination). The task asks you to replace it with a rules-based scheduler that:

- Covers every day with at least 1 doctor per shift slot.
- Respects individual doctor constraints (weekly off, eligible shifts, weekly/monthly caps).
- Handles a gender-restricted shift (OBGYN).
- Lets admins manually override the auto-generated roster.

3. Tech Stack (Strict Requirements)

Layer	Requirement
Backend	Node.js + TypeScript
Database access	Raw SQL only — no ORM (Prisma, Drizzle, TypeORM, Sequelize banned)
Database	Supabase (PostgreSQL) — schema at supabase/duty-doctor-roster-schema.sql
Frontend	React — month calendar / grid UI
AI / LLM	Banned at runtime in the submitted app (AI coding tools like Cursor/Copilot are fine while building)
Deployment	Vercel (frontend/API) + Supabase (DB), public GitHub repo
Scheduling engine	Custom constraint logic in Node — no Python, no OR-Tools

4. Shift Structure

Shift	Timing	Eligibility	Min
Morning	8:00 AM – 2:00 PM	Any doctor	1
Day	10:00 AM – 6:00 PM	Any doctor	1
OBGYN	10:00 AM – 6:00 PM	Female doctors only	1
Afternoon	2:00 PM – 8:00 PM	Any doctor	1
Night	8:00 PM – 8:00 AM	Any doctor	1

Fallback rule: any doctor left unassigned after mandatory allocation must be placed on the Day Shift.

5. Doctors and Individual Rules

Doctor	Allowed shifts	Weekly off	Notes
Dr. Meera Kapoor	All shifts	Wed	Subject to post-night recovery rule
Dr. Rohan Khanna	4 Nights/wk (Mon–Thu only), 1 Morning/wk, 1 Afternoon/wk	Fri	Exempt from recovery rule; consecutive nights allowed
Dr. Aditya Nair	All shifts	Thu	Subject to post-night recovery rule
Dr. Priya Sharma	All shifts	Tue	Subject to post-night recovery rule
Dr. Imran Siddiqui	Day Shift only; max 2 Nights/month	Sun	Subject to post-night recovery rule
Dr. Kavya Menon	All shifts	Sat	Subject to post-night recovery rule

Seed Leave Data — June 2026

These 4 leave days block assignment exactly like a weekly off. They must be visible on the calendar and re-usable when generation reruns.

Doctor	Leave date	ISO date
Dr. Meera Kapoor	Friday, 5 June 2026	2026-06-05
Dr. Aditya Nair	Friday, 12 June 2026	2026-06-12
Dr. Priya Sharma	Friday, 19 June 2026	2026-06-19
Dr. Kavya Menon	Tuesday, 23 June 2026	2026-06-23

6. Special Shift Rules

OBGYN Shift

- Eligible: Dr. Meera Kapoor, Dr. Priya Sharma, Dr. Kavya Menon only.
- Distribute OBGYN duty **equally** among these three eligible doctors.

Night Shift

- Dr. Rohan Khanna: fixed 4 nights/week, Monday–Thursday only, plus 1 Morning + 1 Afternoon shift/week.
- Dr. Imran Siddiqui: max 2 nights/month; all remaining duties go to the Day Shift.
- Remaining nights split **equally** among Dr. Meera Kapoor, Dr. Aditya Nair, Dr. Priya Sharma, Dr. Kavya Menon.

Post-Night Recovery Rule

Applies to everyone **except Dr. Rohan Khanna**. The day after a Night Shift, a doctor can only be given an Afternoon Shift or a day off. Morning, Day, and OBGYN shifts are not allowed the day after a night.

Consecutive Night Shift Rule

Consecutive Night shifts are banned for every doctor except Dr. Rohan Khanna.

Preferred Recovery Pattern

- 1 Day 1: Night Shift
- 2 Day 2: Afternoon Shift or off
- 3 Day 3: Resume normal eligible duties

7. Constraint Priority Order

This is the core logic the generation algorithm must implement — apply constraints in this exact order when assigning shifts:

- 1 Gender restrictions (OBGYN)
- 2 Weekly offs
- 3 Approved leave
- 4 One shift per day
- 5 Maximum 6 shifts per week
- 6 Dr. Rohan Khanna's fixed night allocation
- 7 Dr. Imran Siddiqui's restrictions
- 8 Post-night recovery rule
- 9 No consecutive night shifts
- 10 Equal night shift distribution
- 11 Equal OBGYN duty distribution
- 12 Leftover available doctors → assign to Day Shift

8. Reduced-Staffing Fallback Rules

If too many doctors are unavailable on a given day, shifts are dropped in this order:

Condition	Action	Result
2 doctors unavailable	Drop OBGYN Shift	Keep: Morning, Day, Afternoon, Night
3 doctors unavailable	Drop OBGYN + Day Shift	Keep: Morning, Afternoon, Night

Removal priority (kept longest → dropped first): **Night** → **Morning** → **Afternoon** → **Day** → **OBGYN**.

"Unavailable" includes: weekly off, leave, recovery block, ineligibility for the shift, weekly limit reached, same-day conflict, or a consecutive-night block.

9. General Scheduling Rules

- Roster is generated monthly.
- A doctor can work only one shift per day.
- A doctor can work a maximum of 6 shifts per week.
- Weekly offs are mandatory and cannot be overridden.
- Approved leave blocks all shift assignment for that doctor on that date.

10. What You Must Deliver

- **Supabase schema SQL** — doctors, shift types, monthly rosters, assignments (with manual-override flag), leaves — committed in the repo.
- **Node.js/TypeScript backend** using raw SQL that generates a valid monthly roster and writes it to Supabase.
- **React calendar UI** showing the real, persisted roster (not mock/hard-coded data).
- **Per-day, per-shift editing**: change doctor, clear a slot, or mark a shift inactive (reduced staffing).

- **Manual overrides:** saved immediately to the DB, survive page refresh, shown with a visual indicator, and never silently overwritten by re-generation unless the admin explicitly resets that day/month.
- **"Generate roster" action** per month, with a warning before overwriting non-manual assignments.
- **Validation** on manual edits — toast/error if a rule is broken, with an optional "override anyway + note."
- **README** covering setup, environment variables, the algorithm, and how overrides interact with re-generation.
- **Deployment** on Vercel + Supabase, with a public GitHub repo link.

11. Suggested Build Order

- 1 Supabase schema + seed data (doctors, shifts, leaves) — get raw SQL working first.
- 2 Roster generation algorithm as a pure, testable function — implement the priority-ordered constraint logic.
- 3 API layer (Node/TS, raw pg queries) to persist the generated roster.
- 4 React grid UI to display the persisted roster.
- 5 Manual override CRUD + "don't overwrite manual assignments" logic on regeneration.
- 6 Reduced-staffing fallback logic.
- 7 Validation toasts for rule-breaking manual edits.
- 8 README + deployment (Vercel + Supabase).

Note: staff names are fictional per the original brief and should be used exactly as given, in both seed data and UI.